

Worksheet: Error and Data Analysis - I

Equipment and Setup: Mathematica file – DataAnalysis-I.nb

Section A: Significant Figures

Questions

A-1. Complete the table showing the number of significant digits and the last significant decimal place for each number given.

Number	Number of Significant Digits	Last Significant Decimal Place
3.14159		
590		
590.		
0.0003854		
2.6×10^{21}		

A-2. Give your answers with appropriate significant figures.

$$3.14159 + 0.0003854 = \underline{\hspace{2cm}}$$

$$3.14159 \times 0.0003854 = \underline{\hspace{2cm}}$$

A-3. Using $A = 0.02528$ m, $\omega = 364.6$ s⁻¹ and $t = 0.00243$ s, calculate the result with appropriate significant figures and units. (Note that to evaluate such a function it is essential to use radians, so put your calculator in its radians mode.)

$$A \sin(\omega t) = \underline{\hspace{2cm}}$$

Section B: Target Practice

Computer Setup for Section B

1. The interactive panel shows a target. At the top are controls: there is a Setter Bar with four buttons, one for each trial.
2. Start with Trial 1. Click on the panel to get cross-hairs for aiming. Drag the cross-hairs to the center.
3. Click “Fire” and a dot will appear. Repeat this many times to observe a pattern.

- Repeat the procedure for trials 2 through 4.
- The small button in the upper right resets the panel.

Data Recording for Section B

- Rank the four trials for accuracy and precision in the table below. (Write 1 through 4 into the appropriate boxes.)

Precision ↑		
	Accuracy →	

Questions

B-1. List some factors that could account for a lack of precision in this simulation of target practice.

B-2. List some factors that could account for a lack of accuracy in this simulation.

Section C: Probability Distributions**Computer Setup for Section C.1**

- The interactive panel shows a Setter Bar with two options: “Coin Flip of Heads” and “Dice Rolls of One”.
- Below that is a button “Random” that generates the coin flips or dice rolls. Next to the button there is an Input Field for n , the number of coins or dice. n must be a positive integer.
- The small button in the upper right resets the panel.

Data Recording for Section C.1

1. Start with coin flips and set $n = 10$. For three different trials record the number of heads, x , and the fraction of heads, x/n .

Trial	1	2	3
x			
x/n			

2. Now set $n = 1000$ and similarly record the number of heads, x , and the fraction of heads, x/n for three trials.

Trial	1	2	3
x			
x/n			

3. Now do the same for $n = 1,000,000$.

Trial	1	2	3
x			
x/n			

4. Change the Setter Bar to dice rolls and set $n = 10$. For three different trials record the number of ones, x , and the fraction of ones, x/n .

Trial	1	2	3
x			
x/n			

5. Repeat with $n = 1000$

Trial	1	2	3
x			
x/n			

6. And for $n = 1,000,000$.

Trial	1	2	3
x			
x/n			

Questions

- C-1. The probability of a coin flipping to heads is $1/2$. Discuss what happens to x/n as n increases in steps 1 through 3.
- C-2. The probability of a die rolling one is $1/6$. Discuss what happens to x/n as n increases in steps 4 through 6.

Computer Setup for Section C.2

1. This simple interactive panel has two sliders for the mean μ and the standard deviation σ . The reset button is at the upper right.

Data Recording for Section C.2

1. Adjust the slider for the mean μ . What happens to the graph as μ increases?
2. Adjust the slider for the standard deviation σ . What happens to the graph as σ increases?

Computer Setup for Section C.3

1. There is an Interval Slider that varies the values of x_1 and x_2 . When at the edge of the Interval Slider the values are set to $\pm\infty$.
2. The area between these x -values is listed to the right.
3. A small reset button is at the upper right.

Data Recording for Section C.3

1. For data from a normal distribution, what fraction of the values are less than the mean?

2. What fraction of the values are within one standard deviation σ of the mean? 2σ ? 3σ ?
3. In particle physics they use the 5σ standard. When they are certain of a result within five standard deviations, they are ready to publish their discovery. What is the probability of a normal distribution being more than 5σ from the mean?

Computer Setup for Section C.4

1. There are several controls at the top. At the left there are three tabs: Flip Coins, Binomial Distribution and Sum of Dice Roll. Below each tab are pop-up menus specific to that tab. For each n is the number of coins, trials or dice. For the Binomial Distribution p is the probability of success, where the coin flip is the special case of $p = \frac{1}{2}$.
2. To the right is a Setter Bar, where you choose either a graph of the discrete probability distribution, that same graph with the normal distribution (Gaussian) with the same standard deviation and mean, or a probability table for the distribution.
3. The reset button is at the upper right.

Data Recording for Section C.4

1. What is the probability of flipping 50 coins with exactly 25 heads? (Hint: read this directly off the table.)
2. The variance (the square of the standard deviation σ) for a binomial distribution is $\sigma^2 = n p (1 - p)$. Find the standard deviation for $n = 25$ and $p = 0.8$, and compare with the result from the interactive panel. Now do the same for $n = 60$ and $p = 0.4$.
3. Now click Graph with Gaussian on the Setter Bar. Comment on the comparison between the discrete distribution and the Gaussian as n gets larger.

Section D: Data Analysis**Computer Setup for Section D**

1. The interactive panel generates a random set of data.
2. Clicking “Generate Data” selects a value of μ and σ , the mean and standard deviation of a normal distribution. It then generates 5 random points with that distribution.
3. The reset button is at the upper right.

Data Recording for Section D

1. Click “Generate Data” and record the mean, standard deviation and the data.

	x
$\mu =$ _____	
$\sigma =$ _____	

Questions

D-1. Calculate the statistical mean $\bar{x}$ and the statistical standard deviation s for this data. Show all work.

D-2. The statistical mean $\bar{x}$ and the statistical standard deviation s can be viewed as approximations to μ and σ , the mean and standard deviation of the distribution. Calculate the percent errors in $\bar{x}$ and s taking μ and σ as the accepted values.

% error $\bar{x} =$ _____

% error $s =$ _____